

HR Analysis Dashboard

Business Problems, Insights & Recommended Analysis

Project: Power BI HR Analytics Dashboard | Report: HR Analysis.pbix

1. Executive Summary

This Power BI dashboard provides a management-level view of workforce size, work location, attrition, hiring distribution, employee demographics, department headcount, employee status, and job-title distribution. The dashboard is designed to help HR stakeholders move from a fragmented view of employee data to a single interactive reporting layer.

The current dashboard reports **22K employees**, **75.2% HQ employees**, **24.8% remote employees**, a reported **35% attrition rate**, and an **average age of 38**. These values are from the current dashboard view and can change when filters are applied.

2. Business Problem Statement

HR teams need a centralized way to monitor workforce size, employee retention, hiring patterns, departmental staffing, workforce demographics, and employee status. Without an interactive dashboard, comparing these dimensions can require manual analysis and can make it harder to identify areas that need deeper investigation.

- Limited visibility into workforce distribution across departments and job titles.
- Difficulty monitoring employee attrition and identifying departments with larger termination counts.
- Limited visibility into HQ versus remote workforce composition.
- Historical hiring patterns are available but are not connected to operational context.
- Demographic composition can be monitored, but the current report does not explain causal or operational drivers.
- The dashboard is primarily descriptive; additional employee attributes are needed for root-cause analysis.

3. Business Objectives

- Create a single source of visual HR reporting for management.
- Track workforce KPIs through interactive cards.
- Analyze employee distribution by department, race, and job title.
- Monitor active versus terminated employee composition.
- Understand the historical distribution of employee hiring by year.
- Allow department-level filtering through an interactive slicer.
- Identify business questions that require deeper HR analytics.

4. Current KPI Snapshot

KPI	Current dashboard value	Interpretation
Total Employees	22K	Overall workforce size
Employees in HQ	75.2%	Share of employees working in HQ
Remote Employees	24.8%	Share of employees working remotely

KPI	Current dashboard value	Interpretation
Attrition Rate	35%	Reported attrition metric
Average Age	38	Average employee age

5. Visual-by-Visual Business Insights

Workforce Size

The dashboard reports approximately 22K employees. This establishes the overall workforce scale against which department, location, demographic, and employee-status measures can be compared.

HQ vs Remote Workforce

The dashboard shows 75.2% of employees in HQ locations and 24.8% working remotely. This can support office-capacity planning, hybrid-work analysis, and location-specific workforce planning.

Attrition

The reported attrition rate is 35%. This metric should be investigated further by department, tenure, job title, compensation, location, performance, and exit reason before drawing conclusions about causes.

Department Workforce Distribution

Engineering is the largest department in the displayed view at approximately 6.7K employees, followed by Accounting at approximately 3.3K. Sales and Human Resources are each around 1.8K, while several other departments are below 2K.

Department Terminations

The Attrition by Department visual compares terminated employees by department. In the current dashboard view, Sales displays 335 terminated employees. The full visual should be used interactively to compare all departments.

Hiring Distribution by Year

The Rate of Hire by Year visual displays the percentage distribution of employee records by hire year. Because the visual uses percentage-of-total employee counts, it is more accurately described as the share of employees hired in each year rather than a conventional annual hiring rate.

Race Distribution

The race visual shows workforce composition across demographic categories. White is the largest visible category at 28.49%, followed by other categories shown in the dashboard. This supports demographic reporting but should be interpreted as descriptive workforce composition rather than a measure of hiring or retention drivers.

Active vs Terminated

The donut chart displays approximately 82.31% active and 17.69% terminated employees. These figures should not automatically be treated as the same metric as the 35% attrition rate because the underlying definitions and time frames may differ.

Job Title Distribution

The job-title visual highlights the largest role populations. Visible values include Research Assistant at 754, Business Analyst at 708, Human Resources at 613, a research-related role at 538, and Account Executive at 505.

6. Business Problems Identified

Employee Retention

- The reported attrition rate requires deeper investigation.
- Management needs to know which departments, roles, locations, and employee cohorts experience greater attrition.
- The current dashboard does not expose all root-cause dimensions.

Workforce Concentration

- Engineering and Accounting account for large portions of the displayed workforce.
- Concentration can affect recruitment demand, training capacity, succession planning, and operational risk.

Workplace Location Strategy

- The HQ/remote split can support decisions about office capacity and hybrid-work policies.
- A useful next step is to compare location with attrition, tenure, performance, and job type.

Hiring Pattern Visibility

- Historical hiring distribution is visible, but the dashboard does not explain business events behind changes in hiring volume.
- Additional company and recruitment context would improve interpretation.

Limited Root-Cause Analysis

- The current dashboard primarily answers what is happening.
- Additional fields such as salary, tenure, performance, promotion history, overtime, satisfaction, and exit reason would help explain why patterns occur.

7. Recommended Future Enhancements

Retention Analytics

- Attrition by tenure and salary band.
- Attrition by job title, location, age group, and gender.
- Voluntary versus involuntary termination.
- Exit-reason analysis.
- New-hire retention analysis.

Hiring Analytics

- Monthly or quarterly hiring trend.
- Hiring by department and location.
- New hires versus replacement hires.
- Recruitment-source and funnel analysis.

Workforce Planning

- Headcount forecasting.
- Open-position tracking.
- Employee-to-manager ratio.

- Succession and tenure-risk analysis.
- Skills and capability inventory.

8. Key Business Questions for HR

- Which departments have the highest termination counts and attrition rates?
- Is attrition higher among employees with shorter tenure?
- Do remote and HQ employees show different retention patterns?
- Which job titles have the largest workforce and which have the highest attrition?
- Are hiring patterns aligned with current department workforce needs?
- How does employee age, compensation, performance, or promotion history relate to retention?
- What are the main reasons employees leave?
- Where should HR focus workforce planning and recruitment capacity?

9. Important Metric Interpretation Note

The current dashboard contains both an **Attrition Rate of 35%** and an **Active/Terminated distribution of approximately 82.31%/17.69%**. These numbers should not be assumed to be contradictory or interchangeable. They may use different definitions, denominators, or time periods. For production reporting, document the exact business definition and time window for each metric.

10. Portfolio Value

This project demonstrates practical Data Analyst capabilities in Power BI, including dashboard design, KPI development, DAX-based metrics, interactive filtering, workforce segmentation, visual storytelling, and translation of descriptive data into business questions.

Recommended portfolio positioning: Present this project as an HR analytics case study that moves from workforce reporting toward retention analysis and workforce planning.

11. Limitations

- The current dashboard is primarily descriptive and does not establish causal relationships.
- The available dashboard view does not expose all employee-level attributes required for root-cause analysis.
- KPI values can change with filters, so the current snapshot should not be treated as a fixed organizational total.
- The definition and time window of the 35% attrition metric should be documented before using it for formal HR decisions.